To the Journal of Equine Science

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Subject: New Submission of an Original Article (Full Paper)

I hope this email finds you well. I am writing to submit the following research paper for consideration for publication as a Full Paper in the Journal of Equine Science (JES).

Title of the Manuscript:
Building a model to evaluate cross-industry partners for horse racing IP based on the alignment between fan trends
on social media and company philosophy

1. Background and Objective of the Study

In recent years, while the Japanese horse racing industry has been facing the challenge of an aging core fan base, the rise of racehorse intellectual properties (IPs), such as "UmaMusume Pretty Derby," has triggered new social and economic developments. These include the popularization of equestrian culture among younger generations who previously had little contact with horse racing, and the expansion of support for retired horses by these new, younger demographics. To explore novel methods contributing to customer acquisition in the horse racing industry, I have developed a system model to quantitatively and objectively select and evaluate the optimal cross-industry collaboration partners (e.g., Japan Airlines (Jal)) for horse racing IPs (e.g., "UmaMusume").

2. Implemented System and Demonstration

Conventionally, the selection of cross-industry partners for IPs, not limited to horse racing, has largely depended on the subjectivity and empirical rules of corporate representatives, leaving the process with a challenge of opacity. In this study, I have implemented the following system as an approach to bring objectivity to such black-box decision-making processes. This system evaluates the perspectives of collaboration and the validity of partner selection by analyzing user-generated content (UGC) for both the IP and the candidate partner companies, as well as the corporate philosophies of the IP production company and the candidate partner companies. The system implements the following two analytical methods:

Analytical Method 1: Automatic Collection of SNS Fan Reaction Comments and Visualization via Dashboard

Through the YouTube Data API, system automatically extracted comment data from the official YouTube channels of UmaMusume and Jal, stored them in Google BigQuery, and I built a topic clustering execution environment based on comment contextual similarity using a natural language processing model. Furthermore, by integrating the aggregated results with Tableau, I developed a dashboard that allows for the overview and comparison of the number of likes and replies across the comment groups. As a result, I succeeded in developing a tool that contributes to the discovery and formulation of collaboration perspectives for UmaMusume and Jal that align with public support.

Analytical Method 2: Scoring Function of Affinity Between Collaborating Companies via LLM Prompt Control

To compare the affinity between collaborating companies from the perspective of corporate philosophy (since the production company of UmaMusume is Cygames, this case involves Cygames and Jal), I defined a model to compare and analyze them from three perspectives: "Mission" (What to provide to society/customers, reason for existence), "Value" (Strengths and specific focus in products/services), and "Mind" (Behavioral guidelines and attitudes practiced by employees/teams). Through rigorous prompt control of a Large Language Model (LLM), system automatically extracted the relevant elements from each company's website and constructed model to quantitatively evaluate the affinity between collaborating companies by objectively scoring the similarity of each extracted element.

3. Verification of System Operation through Application of Sample Data

To verify the practicality of the two proposed analytical methods, I applied actual comment data from the official YouTube channels of UmaMusume and Jal as a sample. As a result, I confirmed that the system can visualize the commonalities and differences in the topics of interest shown by both fan bases, as well as quantify the affinity between organizational philosophies.

4. Suitability of this Study for Your Journal

In this project, I aimed to develop a system that can be easily utilized without specialized programming knowledge, primarily targeting staff working in the horse racing industry. By examining strategies to expand the younger demographic of horse racing fans through equestrian IP from the perspectives of both data science and social science, this research represents an interdisciplinary endeavor that aligns with your organization's philosophy of promoting equestrian culture through multifaceted insights. I believe this paper aligns with the aims of your journal, which advocates bridging research and practice, as it brings transparency and objectivity to real-world issues and directly addresses on-site system operations through operational manuals. Furthermore, to ensure the reproducibility of the system proposed in this study, I have explicitly included a technical document (README) detailing the system implementation procedures within the paper and its supplementary materials. This ensures that even researchers outside the field of informatics, as well as industry practitioners, can set up the environment.

5. Declarations

Upon submitting this manuscript, I pledge to comply with your journal's ethical guidelines and submission policies, and I declare the following:

First, this manuscript is an unpublished, original study, and it is not under dual submission or simultaneous review at any other journal.Second, as the sole author, I have confirmed the contents of the final manuscript and the author information.Third, this manuscript has not been pre-registered on any preprint servers or similar platforms.Fourth, I have no conflicts of interest (COI) to disclose in relation to this research.I kindly ask for your review and consideration of this manuscript for publication in your journal.

Sincerely,